NAÏVE BAYES ALGORITHM

1.It is a supervised machine learning machine learning algorithm.
2.It works based on the bayes theorem.
3. Used to solve classification problems.
4. It is most simple and effective algorithm.
5. Develops models which makes quick predictions.
6. It is a probabilistic classifier.
7. Predicts the output based on probability of an object.

Why it is called naïve bayes:

1. It assumes all the features are independent of each and other.
2. If we are adding or changing features it does not make any effect.
3. For example: If the fruit is identified based on the color, shape and taste then the red, spherical and sweet can be identified as apple.
4. Individual feature contributes to identify it is apple without depending on each other.
5. It is called "Bayes" because it works based on principle of bayes theorem.

Bayes Theorem:

1. It is also called Bayes rule or law.
2. It depends on the conditional probability.
3. It determines the probability of hypothesis with prior knowledge.

Formula for bayes theorem:

$P(A/B) = P(B/A)\ P(A)/P(B)$

P(A/B)-Posterior probability
P(B/A)-likelihood probability
P(A)-prior probability
P(B)-marginal probability

1. **Posterior probability:** probability of hypothesis A on the observed event B.
2. **Likelihood probability:** probability of the evidence given that the probability of hypothesis is true.
3. **Prior probability:** hypothesis before observing the evidence.
4. **Marginal probability**: probability of evidence

Working of Naïve Bayes Classifier:

1. Convert the given dataset into frequency tables.
2. Generate likelihood table by finding the probabilities og given features.
3. Use bayes theorem to calculate the posterior probability.

Numerical on naïve bayes classifier:

Suppose we have a dataset of weather conditions and corresponding target variable "Play". So, using this dataset we need to decide that weather player should play tennis or not on particular day according to the weather conditions.

Problem: If the weather is "Sunny" then the player should play or not.

Solution: To solve this first consider the below given dataset.

SI No	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

1. **Frequency table for the "Weather".**

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	**10**	**4**

2. Likelihood table weather condition.

Weather	No	Yes	
Overcast	0	5	5/14=0.35
Rainy	2	2	4/14=0.29
Sunny	2	3	5/14=0.35
All	**4/14=0.29**	**10/14=0.71**	

3.Applying bayes algorithm

$$P(\text{Yes}|\text{Sunny}) = P(\text{Sunny}|\text{Yes}) * P(\text{yes}) \,|\, P(\text{Sunny})$$

$P(\text{sunny}|\text{yes}) = 3/10 = 0.3$

$P(\text{yes}) = 0.71$

$P(\text{sunny}) = 0.35$

So, $P(\text{Yes}|\text{Sunny}) = 0.3 * 0.71 \,|\, 0.35 = 0.60$

$$\boxed{P(\text{Yes}|\text{Sunny}) = 0.60}$$

$$\boxed{P(No|Sunny) = P(Sunny|No) * P(No) / P(Sunny).}$$

$$P(Sunny|No) = 2/4 = 0.5$$

$$P(No) = 0.29$$

$$P(Sunny) = 0.35$$

So, $P(No|Sunny) = 0.5 * 0.29 / 0.35 = 0.41$

$$\boxed{P(No|Sunny) = 0.41}$$

P(Yes/Sunny)>P(No/Sunny)

As we can say P(Yes/Sunny) is greater than P(No/Sunny)

Therefore the Sunny Day ,The player can play the game.

Advantages:

1. It is widely used for text classification problems.
2. It is used for binary and multi class classification.

Disadvantages:

1. Naïve bayes classifier assumes that all the features are independent or unrelated so it cannot learn relationship between features.

Applications:

1. Credit scoring
2. Medical data classification
3. Sentiment analysis
4. Text classification

Naïve Bayes Classifier for Sentiment Analysis:

TRUE SENTIMENT	TEXT
POSITIVE	The food was good
POSITIVE	The food tasted really good
POSITIVE	The service was good
POSITIVE	The price was reasonable
NEGATIVE	The location was not good

- We will take the data from various sources such as twitter and Facebook.
- Sentiment is nothing but the reviews given by the users or the customers.
- The reviews can be positive, negative or neutral.
- The data will be pre-processed using NLP.
- Then using this data, we will develop a Naïve Bayes Classifier in order to classify between various sentiments.